

Macro Finder

Full-stack nutrition app with personalized macro targets and meal tracking

Role: Full-stack developer (SENG 513 team project) · University team project (course repository is private)

React

Node.js

PostgreSQL

Developed a responsive nutrition platform as part of a university team. It calculates personalized macro targets from goals, activity level, and dietary preferences, and supports ongoing meal logging.

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Personalization Inputs

Goals, activity level, dietary preference

Full

Meal History

Daily logs and historical summaries

Mobile

First UI

Responsive across desktop and mobile

Problem

Generic calculators ignore user-specific constraints and make long-term meal tracking difficult.

Solution

Macro Finder combines flexible profile inputs, personalized calculations, and persistent tracking so users can follow a realistic nutrition plan.

Architecture

- React client for onboarding, macro recommendations, and logs
- Node.js API managing profile and nutrition endpoints
- PostgreSQL schema for users, meals, macro snapshots, and history
- Reusable query patterns for aggregate views over time

Highlights

- Computes user-specific macro distributions with edge-case handling
- Supports meal logging and historical tracking
- Normalized relational schema for aggregation queries
- Mobile-friendly, responsive interface

Challenges and Tradeoffs

- Challenge: Personalized recommendations across varied profiles. Change: Edge-case handling for goal and activity combinations.
- Challenge: Keeping meal history performant. Change: Normalized the schema and optimized aggregate queries.
- Tradeoff: Rich interactions vs simplicity. Decision: Prioritized clarity and speed over heavy UI complexity.

Outcomes

- A practical nutrition workflow instead of one-off calculations
- A data model ready for future recommendation features

- Responsive performance across desktop and mobile

Full case study: <https://jaskeerat-portfolio.vercel.app/projects/macro-finder>